

SAFE WORK METHOD STATEMENT

Persons responsible for supervising, inspecting work, work methods, plant & equipment and tools	Supervisor/Foreman	Date:	
High Risk Construction Work:	INSTALLATION OF CULVERT STRUCTURES – EXCAVATION/TRENCHING/PIPELAYING/BACKFILLING	Project:	
SWMS prepared by Patrick Monaghan - Paneltec Group -Operations Manager and in consultation with management and workers – 2025. SWMS Review Date: August 2026 unless required earlier		Location:	
Signature: _____			
Approval to use this SWMS: “I certify that the attached SWMS has been reviewed by me, is endorsed for compliance with WorkSafe guidelines and is approved for use”			
<u>John Guy</u> Name	<u>Director - Paneltec Pty Ltd</u> Position	<u>0408 449 023</u> 	<u>31/08/2025 – V12.0</u> Date

RISK MATRIX

Level	Description of Consequence or Impact	Consequence	Likelihood / Probability		
			L Likely	M Moderate	U Unlikely
H (1) (High level of harm)	Potential death, permanent disability or major structural failure/damage. Off-site environmental discharge/release not contained and significant long-term environmental harm.	H (1) (High)	1	1	2
M (2) (Medium level of harm)	Potential temporary disability or minor structural failure/damage. On-site environmental discharge/release contained, minor remediation required, short-term environmental harm.	M (2) (Medium)	1	2	3
L (3) (Low level of harm)	Incident that has the potential to cause persons to require first aid. On-site environmental discharge/release immediately contained, minor level clean up with no short-term environmental harm.	L (3) (Low)	2	3	3
Level	Likelihood / Probability				
Likely	Could happen frequently				
Moderate	Could happen occasionally				
Unlikely	May occur only in exceptional circumstances				

HIERARCHY OF RISK CONTROL

The hierarchy of controls must be applied in the order below for all identified hazards.

Level 1 Eliminate the hazard
Level 2 Substitute the hazard with something safer.
Level 3 Isolate the hazard from people. Reduce the risks through engineering controls.
Level 4 Reduce exposure to the hazard using administrative actions.
Level 5 Use personal protective equipment The control measures that you put in place should be reviewed regularly to make sure they work as planned

ACTIVITY ANALYSIS

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
Site Safety Rules General safety rules	Injury to workers/public	1	- Operational flashing lights and reversing beepers on plant. - Road to be Closed during works. - Assess the hazards and risks on site eg steep embankments, trees & narrow roads, implement control measures – record findings on Site Specific Induction & Tool Box Meeting form and hold a toolbox meeting with all workers on site to discuss hazards and control measures. - Only suitably trained and competent workers operate plant and equipment. - Seat belts to be worn at all times.	Supervisor	2
	If workers are not wearing the correct PPE personal injuries can occur.	2	All workers to wear safety vests, safety footwear and head protection and other PPE as deemed necessary.	All	3
	Falling or tripping over unstable ground or obstacles.	2	- Maintain good housekeeping. - Barricade off unsafe areas where practicable. - Clearly mark protruding objects using paint, caps or hazard tape.	All Supervisor Supervisor	3
	Manual handling injuries	2	- Manual handling tasks are to be limited where possible and lifting to be done by mechanical means where appropriate and safe. - Correct manual handling techniques are to be followed at all times. - Workers are to request assistance when they deem an item too heavy to be lifted by one person. - Regular breaks from manual handling tasks- job rotation.	Supervisor All	3
	Falling from steep embankments	1	- Appropriate safety harnesses are to be used as required. - Workers to be instructed in the correct fit and use of safety harnesses prior to using.	Supervisor	2
Setting out Traffic Control; cones and or signs	Being run over by vehicular traffic	1	- Workers wear high visibility safety vests at all times. - Traffic control set out vehicles to display flashing amber lights. - Traffic control provided by client as required.	All Supervisor	2

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
Machine Pre Start Check	Unsafe working conditions. Potential for accident.	1	1. Equipment pre start checklist forms filled out & defects documented, supervisor notified. 2. Ensure windows/ mirrors are clean. 3. Ensure all controls are in neutral or park. 4. Ensure all gear in the cabin is correctly stored. 5. Adjust the seat and use the seat belt.	Operator	2
Machine Starting checks	Unsafe working conditions. Increased likelihood of accident during operation.	1	1. The operator must be seated, belted and in full control of the machine when starting. 2. Check the operation of all of the controls and gauges. 3. Listen and watch for defects. 4. Test hydraulic functions. 5. Ensure all grounded attachments are safely retracted 6. Test lights, horn and travel alarm. 7. Check safety locking pins in quick hitch are locked for buckets or attachments.	Operator	2
STEP 1. Excavate for concrete inlet pit, concrete pipes & concrete end walls. Clearing & grubbing. Excavation placement.	Striking other objects whilst slewing excavator, materials on the ground flicking up as excavator travels over site. Overbalance of machine at reach.	1	1. Be aware of what obstacles are around the site especially when slewing excavator. 2. Assess what weights the machine can handle at reach safely before attempting to load trucks. 3. Do not carry passengers. 4. Follow 'SWMS -15 General excavation and trenching'. 5. Carry the bucket as close to the ground as possible whilst traveling. 6. Travel & operate at a safe speed for the operation being undertaken.	Operator	2
	Attachments or bucket disconnecting from quick hitch.	1	1. Ensure quick hitch safety pins are engaged whilst bucket or attachments are engaged into quick hitch.	Operator	2
	Release of hydraulic hoses when changing work implements.	1	1. Ensure hydraulic pressure is taken off hydraulic hoses before releasing couplings. Pressure is normally released if engine is shut down and work tool pedal is pressed back and forth or two-way flow ball valve is switched off.	Operator	2
	Uncontrolled slippage	1	1. Assess ground surface conditions eg; wet ground conditions, steep slope.	Operator	2

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
			2. When working on top of mound make sure excavator is not taken too close to the edge.		
	Ground failure leading to machine rollover and personnel being crushed	1	1. Only competent operators to use machinery. 2. Operator to complete visual assessment of ground conditions. 3. Do not operate close to an overhang or a deep ditch or excavation the weight could collapse the sides of the trench. 4. Always approach embankments or trenches from a 90-degree angle. 5. When working on a slope determine whether to dump the spoil on the high side or low side of the trench. The extra fill will be carted away from job site and stored at a nominated site close to the job.	Supervisor Operator	2
Step 2: Loading of pits, pipes & end walls at stack site (close to job) Loading onto truck with backhoe & transport to job site.	Sling failure	1	1. The Working Load Limit (WLL) of the machine should be marked on the machine. 2. Do not exceed the machine WLL and work radius limits. 3. Ensure site personnel stand well clear of equipment when loading is taking place.	Operator	2
	Struck by moving load/part	1	1. Check the WLL of the machine and the lifting lugs, slings & chains to ensure the load is correctly slung and balanced and is within the WLL.	Operator	2
	Ground failure	1	1. If there is any sign of instability lower the load immediately.	Operator	2
	Failure during lifting	1	1. When the load is raised keep all people at a safe distance. 2. Do not swing loads over people's heads. 3. Do not leave the load suspended in the air. Lower it as soon as possible. 4. Always lower the load to the ground if leaving the machine. 5. Make allowances for materials when they are wet. They will weigh more than normal. 6. If you do not know the weight of the load, do not lift.	Operator	2

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
			7. Do not reeve or nip slings around buckets as the sling can be cut by the cutting edge or teeth of the bucket. 8. Do not place a sling over a tooth or teeth. 9. Place the lifting hook onto the approved lifting lug normally attached to the quick hitch. 10. The person who slings and directs the load must have a dogman's certificate. 11. Inspect all slings prior to use. Are they serviceable?		
	Pipes slipping off truck	2	1. Pipes are securely tied onto tray of truck.	Truck Driver	3
Step 3: Unload pits, pipes and end walls with excavator	Crushing of plant or personnel by pipes.	1	1. Ensure competent dogman secures load. 2. Use certified lifting chains and shackles. 3. Check lifting capacity and WLL prior to lift. 4. Remove excavation bucket prior to lifting pipes. 5. Do not slew load over personnel. 6. Competent dogman required whilst lifting. 7. Refer JSA – 'Using Powered Earthmoving Equipment as a Crane' if applicable.	Supervisor Operator	2
Step 4: Load truck with crushed rock (backfill material) at stack site, transport to job site.	Overloaded vehicle may make braking difficult on steep narrow roads. Unsafe speeds- potential for vehicle accident.	1	1. Do not overload the vehicle. 2. Travel at safe speed suitable for the road conditions. 3. Cover loads where applicable. 4. Ensure personnel at stack site stand clear of the equipment when loading operations take place. 5. All plant & equipment to have operational flashing lights and reversing beepers.	Operator	2
Step 5: Placing of crushed rock with excavator for bedding of pits, pipes & endwalls.	Personnel struck by back fill.	1	1. Use spotter during placement (when plant & personnel working near each other). Ensure hazards discussed at toolbox meeting. 2. Wear hard hats and safety glasses. 3. Reflective safety vests to be worn at all times.	Supervisor All	2
	Back or muscle injury	2	1. Correct manual handling techniques when using shovels.	Workers	3
	Uneven ground	1	1. Plant operator to be aware of uneven surfaces and if possible ensure level ground prior to plant moving bedding and crushed rock.	Operator	2

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
Step 6: Compaction of crushed rock using motorised vibrating plate.	Potential for manual handling injuries and hearing loss.	1	1. Use correct manual handling techniques. 2. Always keep hands, feet and clothing away from moving parts. 3. Always wear safety boots and hearing protection when operating this equipment. 4. Be aware of surface condition and use special care when working on uneven ground or when compacting coarse material. 5. Never allow bystanders to stand close to the machine while it is being started or operating. 6. Seek assistance when lifting machine onto back of vehicle or use a lifting device. 7. Always operate with all the safety devices in place and in working order.	Operator	2
Step 7: Slinging and placement of pits, pipes & endwalls.	Crushing of plant or personnel by pipes.	1	1. Ensure competent dogman secures load. 2. Use certified lifting chains and shackles. 3. Check lifting capacity and WLL prior to lift. 4. Remove excavation bucket prior to lifting pipes. 5. Do not slew load over personnel. 6. Competent dogman required whilst lifting. 7. Refer JSA - 'Using Powered Earthmoving Equipment as a Crane' if applicable.	Supervisor Operator	2
Step 8: Using elastic filler compound to seal any gaps that are visible around concrete structure.	Potential for exposure to substances.	2	1. Use personal protective equipment (PPE) as recommended in the Safety Data Sheet (SDS).	Operator	3
Step: 9 Backfilling of pits, pipes & endwalls by placing crushed rock with excavator & man handling into place.	Personnel struck by back fill.	1	1. Use spotter during placement (when plant & personnel working near each other). Ensure hazards discussed at toolbox meeting. 2. Wear hard hats and safety glasses. 3. Hi-Vis to be worn at all times. Day or night as appropriate.	Supervisor All	2
	Back or muscle injury	2	1. Correct manual handling techniques when using shovels.	Workers	3

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
	Uneven ground	1	1. Plant operator to be aware of uneven surfaces and if possible ensure level ground prior to plant moving bedding and crushed rock.	Operator	2
Step 10: Compaction of crushed rock using motorized vibrating plate.	Potential for manual handling injuries and hearing loss.	1	1. Use correct manual handling techniques. 2. Always keep hands, feet and clothing away from moving parts. 3. Always wear safety boots and hearing protection when operating this equipment. 4. Be aware of surface condition and use special care when working on uneven ground or when compacting coarse material. 5. Never allow bystanders to stand close to the machine while it is being started or while it is running. 6. Seek assistance when lifting machine onto back of vehicle or use a lifting device. 7. Always operate with all the safety devices in place and in working order.	Operator	2
Step 11: Install concrete/sandbag outlet end walls. Load bags onto tandem trailer using backhoe, transport to each jobsite. Place bags around each end of concrete end walls (man-handle)	Personnel struck by backhoe operation. Potential for manual handling injuries.	2	1. Use spotter during loading as required. 2. Wear hard hats and safety glasses. 3. Use correct manual handling techniques when handling cement/sandbags. Use a two person lift where required.	Supervisor All	3
Environmental issues	Damage to the environment if correct controls are not followed.	2	1. Waste is to be disposed of as per instructions from client or taken from site and disposed of in an environmentally friendly manner. 2. Procedures to protect the environment are to be followed as per the company 'Environmental Management Guidelines' document.	Supervisor All	3

Steps in Activity	Potential Hazards	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class After Controls In Place
Additional safety issues that may arise	Sunburn may cause skin damage.	1	1. Workers to wear long pants, long sleeves, sun hats, sunscreen as per 'Personal protective equipment' procedure.	All	2
	Heat Stress	2	1. Management assess the environment during the day and put in place heat stress management techniques such as ensuring workers drink sufficient cool water, take regular breaks & shade covers provided where practicable.	Supervisor All	3
	Chemical/ fume exposure may cause burns or skin problems.	2	1. Store and handle chemicals in accordance with Safety Data Sheets and the 'Hazardous substances and dangerous goods' procedure. 2. PPE provided and should be worn by workers as per SDS.	Supervisor All	3
	Syringes	1	1. Never run your fingers along or under cables, ledges etc. 2. Use a stick or long tool handle to probe such areas. 3. Do not touch the items. 4. Sharp Objects Collection Kit to be used by suitably trained person. 5. Sharps are to be disposed of as per Dept. of Health requirements. 6. Wear latex or plastic gloves for protection when removing a syringe from the area.	Supervisor All	2
	Sharp edged materials	2	1. Workers instructed that when having to move sharp edge materials always wear gloves.	Supervisor All	3
	Working at heights more than 2 metres	2	1. Workers maintain 3 points of contact when climbing up and down from machinery.	Supervisor All	3
	Snakes, spiders and vermin.	2	1. Workers ensure they take care when stepping in long grass and do not step over logs and tin etc on the ground as these are places where snakes & spiders can be found. 2. Always be prepared. 3. Snake Bite emergency kits are carried in all vehicles. 4. Contact emergency services if needed.	Supervisor All	3

ENVIRONMENTAL RISK ASSESSMENT & CONTROL MEASURES THAT MAY BE REQUIRED ON SITE DEPENDING ON THE ACTIVITY					
Work Activity Risk	Hazard	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class after Controls in place
Emergencies	Accident/ Incident Fire Spill	1	1. Personnel have been advised at toolbox meeting and are prepared in the event of an incident occurring. 2. Emergency numbers available on site. 3. Fire Extinguishers/First Aid Kits/Spill Kits on site.	Supervisor All	2
Air Quality	Exhaust Fumes Dust	2	1. Control exhaust emissions eg. Regular maintenance, replace old machinery, fit appropriate emission control measures. 2. Dampen milled surface with water cart. 3. Use water cart where required.	Supervisor All	3
Bitumen/Prime	During heavy rain, recently placed prime or bitumen could emulsify and wash into surrounding waterways.	1	1. Plan prime or bitumen operation so that it is conducted during fine weather. 2. Have material available to clean up a spill if it occurs and dispose of contaminated material accordingly. 3. Protect storm drains during and after application to prevent pollution - have spill clean-up materials available.	Supervisor All	2
Community Relations	Inconvenience as a result of works/traffic management	2	1. Identify from the list of potentially impacted parties who may be impacted by the work being undertaken. 2. Letter drops could be considered (unless specifically required) to notify residents/businesses etc. about works to be undertaken and if it has an effect on them.	Supervisor All	3
Flora & Fauna	Some works may require the removal of vegetation. Native fauna may be at risk from excavation works	2	1. Assess work area for significant vegetation: size of trees, faunal habitat - define work and exclusion areas using fencing and signage. 2. Machinery is to be thoroughly washed down prior to leaving site where noxious weeds are present.	Supervisor All	3
Heritage & Archaeology	Unmonitored excavation at significant sites can lead to loss or destruction of valuable artifacts/relics, and disturbance of historical sites. including cultural heritage and/or archaeological significance	1	1. Undertake a survey of the site to identify any areas of significance. The client's representative may undertake this. Government bodies may be able to assist. 2. Develop a project specific procedure based on the findings and recommendations. 3. Installation of exclusion fencing and signage if required.	Supervisor All	2

Work Activity Risk	Hazard	Risk Class	Risk Control Measure	Responsible Person/s	Risk Class after Controls in place
Noise Pollution, Community Relations & Vibration	Sensitivity to noise - to public & fauna	1	1. Work undertaken during “normal” hours where possible. 2. Affected residents notified in person or in writing. 3. Plant and equipment regularly serviced & fitted with efficient mufflers. 4. Magnitude of vibration reduced by using smaller plant etc.	Supervisor All	2
Soil Management & Water Quality	Soil needs to be managed to ensure that; there are no off-site impacts. Damage to stockpile areas (not prepared properly). Mud deposited onto roadways.	2	1. Wash down machinery in only nominated areas. 2. Not within 20 metres of water courses. 3. Use only approved and nominated stockpile areas. 4. Build bund wall out of material available on site. 5. Drivers to check vehicle tyres/bodies for build up of mud when leaving stockpile sites and remove if necessary.	Supervisor All	3
Fuels and Chemicals	Contamination of waterways due to spillages of oil, degreasers, diesel, etc.	1	1. Store fuel and chemicals in accordance with relevant standards/guidelines. 2. Minimise storage on site and locate storage facilities away from drains and waterways. 3. All containers clearly labelled and Safety Data Sheets are available on site. 4. All drums stored securely and in a bunded area. 5. Spill kits or alternative spill containment materials available on site. 6. Regular maintenance of plant hoses. 7. Use correct refuelling equipment.	Supervisor All	2
Visual Impact Litter and Amenities Waste Management	Litter such as paper, plastic, and other refuse as well as being a possible safety risk may be carried off-site or washed into waterways resulting in pollution and danger to native fauna. Litter is also visually obtrusive. Production generated lighting, litter, mud, dirt and dust may also impact residents.	2	1. Use site induction to communicate the requirement for a tidy site. Also stress consideration for local residents. 2. If as a result of our works, dirt, mud, or litter affects a resident, then we will rectify this problem immediately. 3. Inspect the site frequently and insist on tidiness. Ensure public roads are free of dirt/mud from the work site. 4. Broken-out paved surfaces shall be repaired as soon as possible. 5. Remove all rubbish from site each day. 6. Waste asphalt will be recycled where possible.	Supervisor All	3

Personal Protective Equipment						
All site personnel must wear the required PPE specified in the controls above, such as:						
Safety Hi-vis Vests/Shirts	Safety Boots	Sunscreen	Broad Brimmed Sun Hat	Sunglasses	Gloves specific to the task	White reflective overalls/night
Hard Hat where required	Safety Glasses	Hearing protection where required	Long clothing	Dust masks	Respiratory equipment if required	
Workers may require Task Specific Training depending on activity each worker will be required to fulfill.						
All workers shall hold a current Construction Induction Card						
- If required Client Site Induction - Activity Induction (SWMS & SWP's)	Vehicle Licences (Car or Truck depending on class of vehicle or mobile plant driven)		Plant Operators licences or competency assessments where required		Competencies for Work Zone Traffic Management	
First Aid certificate training	Manual Handling training		Fire Extinguisher training		Fatigue Management Training	
A Site Specific Induction & Toolbox Meeting will be held each day prior to the start of each job.						
Legislation & Codes of Practice (refer to company Legislation register for applicable Australian Standards applicable to works)						
Work Health & Safety Act 2012 and Work Health & Safety Regulations 2022						
Codes of Practice						
Confined Spaces - 2020	Hazardous Manual Tasks - 2020	Managing Noise & Preventing Hearing Loss at Work-2020		Managing Risks of Plant in the Workplace - 2018		
Construction Work - 2020	How to Manage WH&S Risks - 2018	Managing Risk of Falls at Workplaces - 2020		Managing Electrical Risks in the Workplace -2019		
Excavation Work - 2020	First Aid in the Workplace - 2018	Managing Work Environment & Facilities - 2020		WHS Consultation Co-operation & Co-ordination - 2018		
Welding Processes - 2021	How to Safely Remove Asbestos - 2020	Managing Risk of Hazardous Chemicals - 2018		Safe Design of Structures - 2018		
Plant & Equipment for this activity						
- All plant is inspected prior to sending out to site - Daily maintenance & service checks. - Daily pre-start checks are conducted on all plant and trucks by the operators and daily checklist filled out. - Regular servicing at 250hr intervals or as per manufactures specifications. Service History available on request to the office.						
List of plant for this Activity						
Engineering details/certificates/regulatory approvals/permits/electrical isolation (if applicable):						

Implementation, Monitoring and Review of SWMS

The Supervisor on site has the responsibility to:

- Brief each team member on this SWMS before commencing work.
- Ensure control measures have been implemented prior to the start of works.
- Observe work being carried out.
- If controls documented in SWMS are not adequate, stop the work, review the SWMS, adjust as required and re-brief the team.
- Ensure team knows that work is to immediately stop if the SWMS is not being followed.
- Retain this SWMS on site for the duration of the high-risk construction work.
- All persons involved in these particular tasks on site must sign this document as read and understood.

SWMS attendance sheet — I have been given the opportunity to provide input into the development and review of this SWMS

Name (please print clearly)	Signature	Company	Date

Daily pre-start meetings will be held to ensure all workers are informed of control measures and additional safety hazards & controls to be noted on toolbox form.